Extractive Rationalization and Model Accuracy in Few-Shot Learning

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Abstract

We analyze the effect of extractive rationalization on large language model prediction in a few-shot setting. Specifically, we measure the extent to which model prediction errors are associated rationalization errors with respect to gold-standard human-annotated extractive rationales for five datasets. We perform two ablation studies to investigate when both label prediction and evidence retrieval errors can be attributed to qualities of the relevant evidence. We find that there is a strong empirical relationship between model prediction and model rationalization, and that while it is difficult to modify model rationalization via exemplars, it is easier to manipulate model interpretation of rationalized evidence in this manner, leading to contradictory results in some cases.

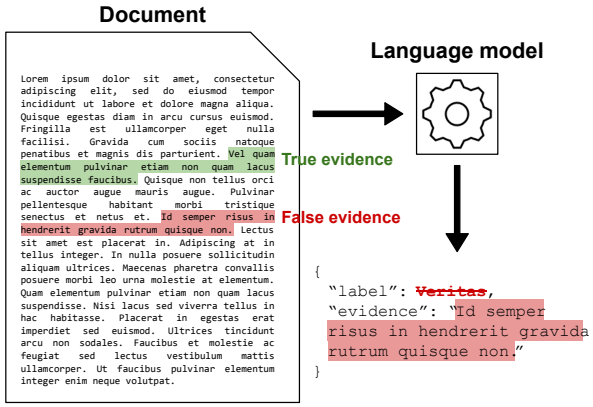


Figure 1: For human or automated verification, we'd like to be able to attribute model prediction mistakes to either misattribution or misinterpretation of relevant within-document evidence.

1 Introduction

The emergence of tremendously powerful large language models (LLMs) like GPT-4 (OpenAI, 2023) and Gemini (Team et al., 2023) has changed the discourse on model interpretability as much as any other aspect of NLP. With only in-context exemplars, these models can not only explain their predictions in fluent, comprehensible and persuasive free text (Zhao et al., 2024), but can also produce verbalized versions of traditional explanations such as token saliency (Feldhus et al., 2023) and extractive rationales (Gurrapu et al., 2023).

The need for model interpretability is at least as urgent as it has ever been. Contemporary LLMs have dramatically lowered barriers to the (seemingly) effective use of NLP for traditionally labor-intensive tasks, leading to a gold rush of LLM applications across a vast range of industries (Zhao et al., 2023b). But these models are prone to failure, meaning that they can't be used for high-stakes tasks such as medicine or legal applications without human verification (Fok and Weld, 2023). And even when they perform well, there are still ethical and often legal reasons that such decisions need to ultimately be (1) reasoned; and (2) attributable to a human decision-maker (Zhao et al., 2024).

In this paper, we focus on **extractive rationales**, i.e. discrete snippets of the input text that the LLM identifies as evidence for its prediction (Lei et al., 2016). As an explanatory mechanism, they lack the expressive power of free-text explanations, but their advantage is that they are directly anchored in the input text. If the goal of interpretability is easier human verification of model predictions (Fok and Weld, 2023), then this concreteness is important because it provides a specific, scrutinizable link between the model prediction and the input text.

However, in order for LM-produced rationales to be useful in verifying model predictions, there are a few conditions that must hold. First the model must be able to reliably extract rationales from the input document; second, there needs to be a causal relationship, or at least strong empirical relationship, between model predictions and model rationales. Finally, it is much easier to verify/correct imprecise extracted evidence than to recover fully missing evidence, so it would be helpful to know which type

of error is more prevalent and a greater driver of prediction error.

We investigate these issues via several variants of forcing a language model to extractively rationalize its predictions in a few-shot setting. We address the following research questions: (1) How does asking the model to (extractively) rationalize its prediction affect its prediction accuracy? (2) To what extent can we attribute label prediction error to rationalization error (with respect to human-annotated rationales)? (3) What types of rationalization error are associated with prediction errors? (4) Under what circumstances does the model fail to retrieve key evidence within the document?

We experiment with five datasets for which gold-standard human-annotated extractive rationales are available, covering a wide range of tasks: MultiRC (Khashabi et al., 2018), SciFact (Wadden et al., 2022), WikiAttack (Carton et al., 2018), Evidence Inference (DeYoung et al., 2020), and HealthFC (Vladika et al., 2024). We find that there is indeed a strong correlation between label and rationale accuracy, which suggests empirical if not by-construction faithfulness in LLM-generated extractive rationales. We find that, insofar as we can attribute model model mistakes to rationalization errors, they stem more from misinterpretation of correct and relevant evidence than missing evidence, and that the model is likely to miss evidence in cases where other viable evidence is available.

2 Related work

LLM Explanations. Extractive rationales are a feature attribution-style explanation method which seeks to identify the subset of tokens with a text responsible for a model's prediction on that text (Lei et al., 2016). Past approaches have used either a two-stage "explain-then-predict" setup where the prediction is generated from the extracted rationale, or an "explain-and-predict" setup where the rationale and prediction are generated concurrently (Zhang et al., 2021). The advantage of the former is that it guarantees *faithfulness* (Wiegreffe and Marasović, 2021) by construction, where the prediction can be positively attributed to the rationale.

Because of the ability of contemporary LLMs to generate fluent and comprehensible text, recent explainability work has focused largely on free-text abstractive rationalization, which is more expressively powerful than extractive rationales and can be very persuasive (Huang et al., 2023; Si et al., 2023; Zhao et al., 2024). However, free-text explanations are particularly vulnerable to faithfulness problems (Agarwal et al., 2024; Lyu et al., 2024), leading some recent work to continue using LLMs in an extractive manner (Ludan et al., 2023; Malaviya et al., 2023), sometimes in combination with abstractive free-text (Majumder et al., 2022).

Learning with intermediate reasoning. The idea of explaining LLM predictions is related to the idea of prompting them to generate intermediate reasoning for their predictions (such as Chain-of-thought prompting (Wei et al., 2022)), and of verifying that intermediate output to correct errors (such as maieutic prompting (Jung et al., 2022), verify-and-edit (Zhao et al., 2023a), and chain-of-verification (Dhuliawala et al., 2023)). However, these intermediate outputs can also suffer from faithfulness problems (Turpin et al., 2023). Also related is the concept of "learning from explanation", or using human-generated explanations as additional training signal for models (Hase and Bansal, 2021; Carton et al., 2022). At the extreme end of the spectrum is work on prompting LLMs to perform conceptual reasoning, which often operates over concepts derived extractively from the input text (Stacey et al., 2023; Ludan et al., 2023).

Needle-in-the-haystack evaluation. Evidence recall is particularly important in rationalization contexts, where the correctness of rationales is necessary for a model's response to be useful and verifiable. One approach for testing the in-context retrieval ability of long context LLMs is the "needle in a haystack" test where, in a random statement is placed in the middle of a long context window and the model is asked to retrieve this statement (Dhinakaran and Jolly, 2024). It is posited that as a document's size increases, it becomes increasingly difficult for the model to retrieve the relevant portions of text. Additionally, the location of the relevant statement(s) plays a role in the model's recall ability.

3 Datasets

We conduct our analysis on five datasets for which human-annotated extractive rationales are available, covering a diverse range of tasks: MultiRC (Khashabi et al., 2018), SciFact (Wadden et al., 2022), and WikiAttack (Carton et al., 2018), Evidence Inference (DeYoung et al., 2020), and HealthFC (Vladika et al., 2024). To reduce API

access costs, we perform our analysis on randomly-sampled 300-item subsets of the development set for each dataset except WikiAttack. The development set for WikiAttack consists of only 250 items, thus subsampling is unnecessary.

SciFact (Wadden et al., 2022) is a scientific claim verification dataset, facilitating the task of selecting abstracts from the research literature which either support or refute a given scientific claim. The dataset contains 1,400 expert-written claims, paired with evidence-containing abstracts annotated with veracity labels and sentence-level rationales. Rationales are direct spans of text from the document which support the class label given. Each claim has a class label of *SUPPORT*, *NO INFO*, or *CONTRADICT*.

MultiRC (Khashabi et al., 2018) is a reading comprehension dataset containing sets of short paragraphs and questions that depend on information from multiple sentences within the paragraph. Each paragraph-question pair contains five answers, with a variable number of correct answer-options. Additionally, answer-options do not have to be a span from the text. The dataset contains sentence-level rationales in the form of relevant sections of the paragraph, as well as *True* or *False* labels for each answer candidate.

WikiAttack (Carton et al., 2018) contains excerpts of Wikipedia comment threads original included in the WikiToxic dataset of (Wulczyn et al., 2017) which are scanned for instances of personal attacks or harassment. Class labels are given as either *attack* or *nonattack*, and rationales for comments labeled as attack are given in the form of spans from the comment.

Evidence Inference (DeYoung et al., 2020) dataset contains scientific articles and queries asking to compare the relative effectiveness of two treatments with respect to a given outcome. The label domain is *significantly increased*, *significantly decreased*, and *no significant difference*. Annotations are in the form of quotes from the text which support the label.

HealthFC (Vladika et al., 2024) is a medical domain QA dataset. Given a health-related claim, verdicts are decided from a systematic review or clinical trial document, as one of *supported*, *refuted*, and *not enough information*. Additionally, annotations containing up to five quotes from the document are provided.

After reviewing the dataset, we found discrepancies between the original, expert-annotated, German verdicts, and their English label mappings. Thus, we performed a re-labeling of the dataset, by first asking GPT-4 to provide an English label given the query and German verdict, and then by manually labeling each item in the dataset, using a holistic review of the original English label, German verdict, cited evidence, and full document text.

A) Rationalized prediction prompt

```
Based on the document between the <doc> tags,
classify the claim between the <claim> tags as
SUPPORT, CONTRADICT, or NOINFO, with respect to
the document. Respond in json format, with a
'label' field for the classification and an
'explanation' field listing the parts of the
document that support the label.
```

{{NOINFO, CONTRADICT, and SUPPORT exemplars}}

<doc>**Genetically identical cells sharing an environment can display markedly different phenotypes**... allows a trial commitment to multicellularity that external signals could extend.</doc>
<claim>**Gene expression does not vary appreciably across genetically identical cells.**</claim>

```
{"label":"NOINFO", "explanation":[]}
```

B) Evidence-only prompt

{{Prompt instructions}}

<doc>Genetically identical cells sharing an environment can display markedly different phenotypes.</doc>
<claim>Gene expression does not vary appreciably across genetically identical cells.</claim>

```
{"label": "CONTRADICT"}
```

Figure 2: Two GPT-4 prompts for the same SciFact item. Claims and true evidence (i.e. human-annotated rationale) underlined for emphasis. Model output in green. (A) The model misses true evidence and misclassifies claim with full prompt and random exemplars. (B) The model chooses the correct class when presented with only the true evidence.

4 Experimental design

We investigate (1) how forcing an LLM to extractively rationalize its prediction affects its performance, and (2) the relationship between rationalization and prediction error.

Some rationalization error seem to be attributable to the model being unable to recognize

the within-document evidence (in the form of the human-annotated rationale) as being relevant to the claim. Fig. 2A shows an example, where GPT-4 is unable to recognize evidence referring to "phenotypes" as being relevant to a claim concerning "gene expression".

Experimental conditions. Our full experiment includes six few-shot learning conditions, and one zero-shot condition: (1) zero-shot for label prediction; (2) few-shot for label prediction; (3) explain-then-predict for label and rationale prediction; (4) predict-then-explain for label and rationale prediction; (5) ground-truth evidence provided for label prediction; (6) ground-truth evidence providied without the document text for label prediction; (7) ground-truth evidence redacted from the document for label prediction.

5 Implementation details

The *gpt-4-0613* version of GPT-4 is used for all experiments for MultiRC, WikiAttack, and SciFact. In order to mitigate issues with larger context lengths, *gpt-4o-2024-05-13* is used for Evidence Inference, and *gpt-4-turbo* for HealthFC. All experiments are conducted via the OpenAI API, with temperature set to the default value of 0.7.

Rationalized prediction. Figure 2A illustrates the prompt used to perform rationalized prediction. We produce predictions and extractive rationales concurrently in an "explain-and-predict" manner, introducing the possibility of unfaithful rationales with no actual relationship to the model's label prediction. Part of our analysis examines this relationship empirically. The prompt presents sampled exemplars in JSON format, and requests model output in this format.

Exemplar sampling. All few-shot conditions involve sampling two exemplars for each possible class, resulting in 4-shot learning for MultiRC and WikiAttack, and 6-shot learning for SciFact. We randomly sample exemplars, and shuffle exemplar order, separately and independently for each item. For MultiRC and SciFact, exemplars are sampled from the training set. For WikiAttack, where human rationales are unavailable for the training set, they are instead sampled from elsewhere in the development set, resulting in effectively 250-fold cross-validation.

For experimental conditions 3 through 6, exemplars are randomly sampled solely from those whose true evidence (human-annotated rationale) meet the required criteria (low versus high claim-evidence similarity and frontloaded versus backloaded evidence position, respectively).

6 Results

6.1 Rationale vs label performance

Table 1 shows the mean rationalization performance (compared to human-annotated rationales) for correct and incorrect predictions made by the model when using fully random exemplars. Figure 3 illustrates the distribution of rationale F1 under these conditions.

What we observe is that there is indeed a significant correlation between rationale performance and label accuracy. This effect is strong for SciFact, WikiAttack, and Evidence Inference in all cases, with respective rationale F1 differences of 0.65, 0.37, and 0.33. For SciFact, hardly any incorrect predictions display any level of correct rationalization. In MultiRC, there is a significant difference in the rationale F1 and precision (differences of 0.22 and 0.20, resp.), however not in the recall. Lastly, we observe no significant differences in the HealthFC partition.

This result suggests that there is a strong empirical relationship between identifying correct evidence (in the form of extractive rationales) and producing correct predictions. It suggests that if rationalization performance could be improved, then so too could prediction performance, which we evaluate in the following sections.

6.1.1 Effect of rationalization order

In Table 2, we compare asking the model to first rationalize its prediction and then provide a label, versus rationalizing after making its label prediction. In MultiRC, SciFact, and HealthFC, changing the request order caused a statistically significant difference in the label prediction performance. With the exception of evidence inference, this difference is mostly in favor of the predict-then-explain condition, suggesting that in most cases constraining the model to pre-emptively justify its label has a slightly negative impact on its label accuracy.

6.2 Explaining label and rationalization errors

In order to more fully explore the effect of imposed rationalization and the relationship between rationalization and prediction performance, we evaluate

Dataset	Correct predictions				Incorrect predictions			
	Count	Rationale			Count	Rationale		
		F1	Recall	Prec.		F1	Recall	Prec.
MultiRC	253	0.71	0.68	0.83	41	0.59**	0.62	0.63***
SciFact	249	0.76	0.77	0.82	50	0.11***	0.13***	0.12***
WikiAttack	248	0.73	0.81	0.72	52	0.36***	0.45***	0.34***
Evidence Inference	159	0.61	0.62	0.69	20	0.38***	0.32***	0.48**
HealthFC	236	0.31	0.32	0.37	64	0.25	0.25	0.31

Table 1: Rationalization performance for correct vs incorrect label predictions in the few-shot learning with rationales condition. Asterisks denote significant differences between values using a 2-sided t-test; *:p<0.1; **:p<0.05; ***p<0.005

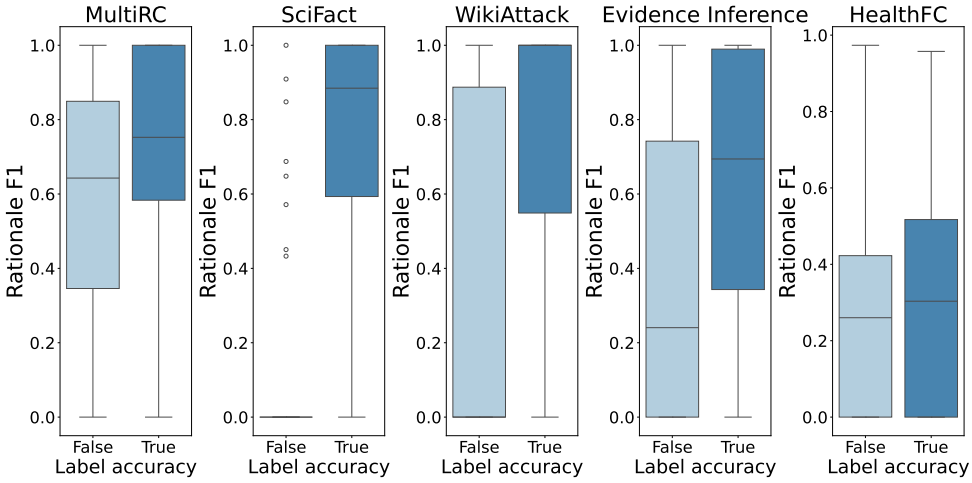


Figure 3: Box-and-whisker plots of label prediction error versus mean predicted rationale F1 for all three datasets.

the model in several additional prompting conditions: (1) **zero-shot**, with neither exemplars nor rationalization; (2) standard **few-shot** with no rationalization; (3) **explanation given**, in which the human-annotated rationale is provided as part of the input prompt; (4) **explanation given without document**, where the human annotated rationale is provided without the context of the surrounding document; and (5) **evidence occluded**, where the human-annotated rationale is removed without replacement from the document.

Table 3 summarizes the basic label prediction accuracy results. For two datasets, Evidence Inference and HealthFC, forcing the model to rationalize its prediction has a substantial positive effect (+7.9 and +4.3 respectively). The effect is marginal for the other three. SciFact, Evidence Inference, and HealthFC see an increase in performance when provided with the ground-truth rationales, with no improvement for WikiAttack and MultiRC. Furthermore, we note a uniformly sharp decrease in performance across all datasets when the human-annotated evidence is redacted from the input document.

So we see mixed results across datasets from the effect of both forcing the model to extractively rationalize its predictions, and providing human-annotated rationales to the model in the input prompt. However, comparing performance across these conditions allows us to begin understanding the underlying cause of model prediction and evidence retrieval errors.

6.2.1 Why does the model make prediction errors?

When the model makes prediction errors, to what extent can we attribute these errors to missing evidence or misinterpretation of correct evidence? Is it more common for the model to miss key evidence entirely (which is difficult to correct via prompt engineering), or to misinterpret relevant evidence (which can be addressed by reinspection approaches)?

Table 4 explores these questions in the form of a contingency table of label prediction errors in

Dataset	Prompt Structure	Label acc	Rtnl. F1	Rtnl. precision	Rtnl. recall
MultiRC*	Predict then explain	0.897	0.676	0.794	0.646
	Explain then predict	0.861	0.692	0.804	0.667
SciFact*	Predict then explain	0.870	0.666	0.737	0.664
	Explain then predict	0.833	0.650	0.705	0.664
WikiAttack	Predict then explain	0.840	0.709	0.694	0.784
	Explain then predict	0.827	0.670	0.653	0.748
Evidence Inference	Predict then explain	0.870	0.584	0.627	0.633
	Explain then predict	0.888	0.588	0.662	0.587
HealthFC*	Predict then explain	0.834	0.343	0.433	0.326
	Explain then predict	0.787	0.297	0.357	0.306

Table 2: Label and rationale performance for different prompting paradigms. Asterisks denote significant differences in label predictions between the prompt conditions using the McNemar test; *:$p<0.05$

the explain-then-predict condition. Label errors in this condition are divided two ways: (1) low- vs. high rationale recall within this condition, indicating where label errors coincided with missing versus recovered human-annotated evidence; and (2) correct versus incorrect predictions in the explanation-given condition, representing cases where the model *would* have made the correct prediction if only the human-annotated evidence had been provided to it.

In this division, the (low-recall, correct label) condition represents cases where the model's label prediction mistake would have been overturned by presenting it with the human rationale, meaning that we can attribute the mistake to the missing evidence. The (low-recall, incorrect label) condition represents cases where the model's mistake persists with or without the human rationale, making it impossible to attribute. The (high-recall, correct label) condition implies that there was extra erroneous evidence the model picked up in addition to recovering the human rationale, which caused its label prediction error. Finally, the (high-recall, incorrect label) condition represents scenarios where the model successfully recovered the human rationale, but made the incorrect prediction regardless (misinterpretation).

What we find is that in three of out five datasets (MultiRC, SciFact, and WikiAttack), label prediction errors are mostly associated with high rationale recall (misinterpretation and extraneous evidence) rather than low recall (missing evidence). In MultiRC and WikiAttack, the model is most likely to recover the human-annotated evidence but be unable to interpret it properly, while for SciFact the model is likely to recover both the human rationale and extraneous evidence which causes it to produce an incorrect prediction.

For two datasets, HealthFC and Evidence Inference, a majority of prediction errors are associated with low rationale recall, but only or HealthFC would providing human rationales to the model produce a correct label. This means that HealthFC is the only dataset to for which we can attribute a majority of errors to missing key evidence that the model did have the capacity to interpret, the scenario investigated by needle-in-the-haystack evaluations. In other cases, the model is more likely to recover the relevant evidence and then misinterpret it, which is a hopeful sign for interpretable systems based on extractive evidence.

6.2.2 Why does the model miss key evidence?

Similar to section 6.2.1, we can use the results of different prompting paradigms to ask why the model misses key evidence. In particular, we can investigate the following hypothesis: **that the model tends to miss evidence primarily when it is unable to interpret it correctly.**

Table 5 represents a contingency table of low-human-rationale-recall instances from the explain-then-predict condition, divided by label prediction error in the explanation-occluded condition and the explanation-given condition. The (occluded incorrect, given correct) contingency represents cases where the human rationale was both necessary and sufficient for the model to correctly interpret and predict the label, while (occluded incorrect, given incorrect label) represents cases where the human rationale was not unnecessary, but also not sufficient to predict the label. Finally, either contingency where the label prediction was correct for the explanation-occluded condition is one where there

Dataset	Condition	Label acc
MultiRC	Zero-shot	0.887
	Few-shot	0.880
	Explain then predict	0.861
	Explanation given	0.887
	Explanation given w/o document	0.827
	Evidence occluded	0.663
SciFact	Zero-shot	0.853
	Few-shot	0.857
	Explain then predict	0.833
	Explanation given	0.940
	Explanation given w/o document	0.907
	Evidence occluded	0.633
WikiAttack	Zero-shot	0.743
	Few-shot	0.840
	Explain then predict	0.827
	Explanation given	0.823
	Explanation given w/o document	0.917
	Evidence occluded	0.533
Evidence Inference	Zero-shot	0.832
	Few-shot	0.809
	Explain then predict	0.888
	Explanation given	0.907
	Explanation given w/o document	0.920
	Evidence occluded	0.689
HealthFC	Zero-shot	0.763
	Few-shot	0.733
	Explain then predict	0.787
	Explanation given	0.947
	Explanation given w/o document	0.850
	Evidence occluded	0.689

Table 3: Label Accuracy for different prompting paradigms

was additional viable evidence beyond the human-annotated rationale, rendering it unnecessary, and low rationale recall much less meaningful.

If (meaningful) misses of human-annotated rationales were mostly associated with them being uninterpretable by the model, a majority of cases would fall in the (occluded incorrect, given incorrect) contingency. Instead, Table 5 shows that a majority of cases in four out of five datasets fall into the (occluded correct, given correct) contingency, meaning that the model mostly fails to retrieve human-annotated evidence when there is additional viable evidence that the model can use to successfully predict the label.

Even when no such additional viable evidence is possible (incorrect label in the occluded condition), a majority of cases for all datasets are ones where the label is correct in the explanation-given condition, meaning these are cases where the model does know how to correctly interpret the key evidence, it simply fails to retrieve it in the explain-then-predict condition. In WikiAttack alone does this contingency represent a majority of overall cases. Hence, the hypothesis that retrieval failures are associated with evidence the model struggles to interpret correctly is not supported in our analysis.

7 Discussion

In order to be able to use extractive rationales to scrutinize and critique the reasoning by which large language models make their predictions, we first need to establish that there is a relationship between model rationalization and model prediction–that the predictions made by the model are conditioned on the rationales it identifies within the input text. The results shown in Section 6.1 do support this hypothesis to a certain extent, showing a strong relationship between model prediction accuracy and model rationalization accuracy relative to a human-annotated gold standard, for four out of five datasets we investigate.

The analysis performed in 6.2 shows that prediction errors are more likely to be a result of imprecise rather than incomplete retrieval of evidence. This is a hopeful sign for any system seeking to allow human verification or automated calibration based on the model-identified evidence. It is easier for prompt engineering approaches to place further scrutiny on intermediate evidence (e.g. chain-of-verification (Dhuliawala et al., 2023)) than it is to recover entirely missing evidence, so it is help-

Explain-then-predict	**Low rationale recall**		**High rationale recall**	
Explanation-given	**Correct label**	**Incorrect label**	**Correct label**	**Incorrect label**
Note	Missing evidence	Unexplainable	Extra evidence	Misinterpretation
MultiRC	0.12 (5)	0.24 (10)	0.22 (9)	**0.41 (17)**
SciFact	0.10 (5)	0.14 (7)	**0.60 (30)**	0.16 (8)
WikiAttack	0.00 (0)	0.06 (5)	0.34 (26)	**0.60 (46)**
HealthFC	**0.73 (40)**	0.07 (4)	0.13 (7)	0.07 (4)
Evidence Inference	0.26 (5)	**0.32 (6)**	0.16 (3)	0.26 (5)

Table 4: Contingency table of label prediction errors in explain-then-prediction condition; split by low vs. high rationale recall and correct vs. incorrect prediction in explanation given condition.

Human rationale occluded	**Label incorrect**		**Label correct**	
Explanation given	**Label correct**	**Label incorrect**	**Label correct**	**Label incorrect**
Note	Interpretable	Uninterpretable	Other viable evidence present	
MultiRC	0.22 (19)	0.14 (12)	**0.60 (52)**	0.05 (4)
SciFact	0.37 (25)	0.10 (7)	**0.51 (34)**	0.01 (1)
WikiAttack	**0.47 (7)**	0.20 (3)	0.13 (2)	0.20 (3)
HealthFC	0.30 (45)	0.01 (2)	**0.65 (97)**	0.04 (6)
Evidence Inference	0.16 (8)	0.08 (4)	**0.76 (38)**	0.00 (0)

Table 5: Contingency table of low human rationale recall instances in explain-then-prediction condition; split by correct vs. incorrect label prediction in explanation-occluded and explanation-given conditions.

ful that the latter is the characteristic error we observe. We also show that when evidence is missed, it is most often when there is other viable evidence present which is equally sufficient for predicting the correct correct label.

Taken together, these results suggest that extractive rationales are an effective way of examining LM behavior in a few-shot setting, and that contemporary LMs are effective at retrieving relevant in-context evidence and interpreting it to produce correct predictions.

8 Conclusion

In this work, we investigate the relationship between prediction and extractive rationalization in few-shot learning. We find that there is a strong correlation between the classification accuracy and agreement with human-annotated rationales. Furthermore, we find that significantly more model error is attributable to imprecise rationalization than incomplete rationalization, which is a positive sign for downstream methods based on this mechanism.

Limitations

A major limitation of this work is that it probes the behavior of a single large language model on a small selection of datasets for which human-annotated rationales are available, so any conclusions we reach are necessarily preliminary and would need to be replicated and extended to achieve a more general validity. Another limitation is that there is no guarantee of *faithfulness* in rationales produced by a LM in a few-shot setting, so we hesitate to claim firm causal links between rationalization and prediction behavior in the model.

Ethical considerations

The risks associated with the work presented in this paper are relatively minimal, as we simply present empirical observational results about the prediction behavior of language models in several settings. Perhaps the main risk is the implication that extractive rationales can be treated as strictly causal of predictions produced by the LM, and so we reiterate that our observations are empirical and based on statistical association.

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A Appendix

The following figures demonstrate the characteristic errors experienced within datasets, using real dataset prompts and model predictions.

Dataset: `WikiAttack`
Query: `Ratings===\n\nHow about something about this dude's MONSTER ratings?\nHe routinely DOUBLES his ENTIRE competition....COMBINED!\n\nhttp://www.mediabistro.com/tvnewser/ratings/the_scoreboard_friday_june_12_118993.asp\n\n Wow! Makes a liberal want to puke in their mouth a little, huh? haha\n\n \n\n===`

Ground-truth Evidence: `[]`
Model's Predicted Evidence: `Makes a liberal want to puke in their mouth a little, huh?`

(True label, Predicted label): `(nonattack, attack)`

Characteristic Error: `Overly cautious, misattributing other forms of negative statements as personal attacks`

Figure 4: WikiAttack errors commonly arise due to the model misattributing any negative statements as a personal attack

Dataset: `MultiRC`
Query: `At what level of force does the table oppose gravity ? || Slightly above than 20 newtons`

Ground-truth Evidence:
`At the same time, the table pushes the book upward. , Its force is also 20 newtons. , The table opposes the pull of gravity.`
Model's Predicted Evidence:
`A book is resting on a table., Gravity pulls the book downward., It has a force of 20 newtons., At the same time, the table pushes the book upward., Its force is also 20 newtons., The table opposes the pull of gravity.`

(True label, Predicted label): `(True, False)`

Characteristic Error: `Annotator Error`

Figure 5: MultiRC errors commonly arise due to annotator errors

Dataset: Evidence Inference
Query: With respect to Cleavage and blastocyst rates, what is the reported difference between VS1 group : COCs were exposed to vitrification solution 1 (VS1) containing 7.5 % ethylene glycol [EG] + 7.5 % dimethyl sulfoxide [DMSO] + 20 % calf serum [CS] in TCM-199 at 37 C for 5 min ; VS1 + VS2 group : COCs were exposed to VS1 for 5 min followed by VS2 (15 % EG + 15 % DMSO + 17.1 % sucrose + 20 % CS) at 37 C for 45-60 sec; and Control group: no treatment?

Ground-truth Evidence:
Cleavage and blastocyst rates did not differ among Control , VS1 and VS1 + VS2 groups in either experiment .
Model's Predicted Evidence:
both cleavage and blastocyst rates were lower (P < 0.001) in the Vitrified group than in the Control,

(True label, Predicted label): (significantly decreased, no significant difference)

Characteristic Error: Misretrieval of incorrect statements that use key query terms

Figure 6: Evidence Inference errors commonly arise due to misretrieval

Dataset: HealthFC
Query: Can hyposensitization by means of drops or tablets administered under the tongue help with hay fever?

Ground-truth Evidence:
Moderate relief without guarantee of success The summarized results of previous studies show that hyposensitization by means of pollen extracts taken under the tongue probably alleviates the symptoms somewhat [1].
Model's Predicted Evidence:
The summarized results of previous studies show that hyposensitization by means of pollen extracts taken under the tongue probably alleviates the symptoms somewhat., On a scale of 0 (no complaints) to 6 (extreme complaints), the allergy symptoms are expected to improve by 0.8 points according to the studies., There is no guarantee of success, these results are merely averages. In some patients, the treatment does not work at all, while it can significantly relieve the discomfort in others.

(True label, Predicted label): (supported, supported)

Characteristic Error: Predicted evidence correct, but doesn't match ground-truth evidence, for a correct label prediction

Figure 7: HealthFC rationale errors commonly arise due to redundant evidence